Martin Chalela

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Research Software Engineer
PhD in Astronomy

github.com/mchalela
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Research Software Engineer with a PhD in Astronomy and over 10 years of experience building scientific Python software. Most recently, for the University of Chicago, I developed the data reduction pipeline for MAROON-X, the high-precision radial velocity spectrograph at Gemini Observatory, porting a legacy pipeline into the DRAGONS framework with full test coverage, from 2D image processing to echelle extraction and wavelength calibration. I lead and contribute to open-source astronomy packages, three of them described in refereed journals (*Astronomy & Computing*, *MNRAS*), and have co-authored 9 refereed papers, 3 as first author. I bring a scientist's understanding of the data together with solid engineering practice (testing, CI, documentation and AI-assisted development), and I taught a postgraduate course on software design for scientists at the National University of Córdoba.

Skills

Tools and Languages	Python, pytest, Git, SQL, Tox/Nox, Docker, LaTeX, Sphinx/Read the Docs, Fortran (basic), C (basic)
Python Scientific Stack	Astropy, NumPy, SciPy, pandas, Matplotlib, DRAGONS, AstroData, Numba, Joblib, Jupyter
Software Development	Design, Testing, CI/CD (GitHub Actions), Packaging, Profiling/Optimization, Documentation, AI-Assisted Development
Data Processing	Data Pipelines, FITS/HDF5, Image & Signal Processing, Spectroscopy, Data Analysis & Visualization
Scientific Research	Bayesian Statistics, Monte Carlo Simulations, Astrophysical Modeling, Academic Writing
Communication	English (fluent), Spanish (native)

Education

PhD in Astronomy — **April 2015 — March 2020**
FAMAF (Facultad de Matemática, Astronomía, Física y Computación), UNC, Argentina
Thesis title: Analysis of Compact Groups of Galaxies using Weak Gravitational Lensing

Bachelor in Astronomy — **March 2007 — March 2015**
FCEFN (Facultad de Ciencias Exactas, Fisicas y Naturales), UNSJ, Argentina

Technical Experience

Scientific Software Developer — **November 2024 — September 2026**
University of Chicago (remote contractor)

- Developed the data reduction pipeline for MAROON-X, a high-precision radial velocity spectrograph at Gemini North, within the Gemini DRAGONS framework, working closely with the University of Chicago instrument team.
- Ported a legacy HDF5-based pipeline to a FITS/AstroData architecture with full test coverage: 2D image processing, echelle spectral extraction, wavelength calibration and SQL database integration.
- Delivered the pipeline to Gemini Observatory; it is now in use by the instrument team for MAROON-X data reduction.
- Repository: github.com/GeminiDRSoftware/MAROONXDR

Data Scientist & Python Developer — **March 2024 — August 2024**
IThreex Global, Argentina

- Developed and deployed machine learning models using scikit-learn and the Python scientific stack into production environments to deliver data-driven value to business clients.
- Containerized models with Docker and collaborated with the DevOps team through GitLab and Slack.

Freelance Software Developer — **April 2023 — March 2024**
Upwork

- Completed multiple independent contracts involving Python development, data analysis, code optimization, and astrophysics consulting, maintaining a 5-star rating across all reviewed projects.

Lead Developer of pytest-deck — **August 2026 — Present**
Open Source Project

- pytest-deck is a pytest plugin built to help developers quickly understand and verify code they did not write themselves, including AI-generated code and tests.
- Serves an interactive web dashboard for the test suite: collect it as a foldable tree, select tests, and watch results stream in live, with tracebacks and coverage. Integrates with the pytest plugin ecosystem.
- Built a FastAPI and Server-Sent Events back-end with a Svelte front-end; every run is an isolated pytest subprocess, so results always match a command-line run.
- Repository: github.com/mchalela/pytest-deck

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Lead Developer of GriSPy **August 2019 — Present**

Open Source Project @ IATE, CONICET

- GriSPy: Grid Search in Python, is a regular grid search algorithm for quick nearest-neighbor lookup focused on optimizing query times.
- As the Lead Developer my tasks are to design, develop, test and document new features.
- I also act as Project Manager with a team of 5 developers.
- Repository: github.com/mchalela/GriSPy

Developer of Nirdust **August 2020 — Present**

Open Source Project @ IATE, CONICET

- Nirdust is a python package that uses near-infrared K-band (2.2 micrometers) spectra to measure the temperature of the dust heated by an Active Galactic Nuclei (AGN) accretion disk.
- As a Developer my tasks are to implement parametric model optimization functionalities as well as testing and documentation.
- Repository: github.com/Gaiana/nirdust

Developer of GARPAR **August 2021 — December 2021**

Open Source Project @ IATE, CONICET

- GARPAR is a Python project (in development) for the generation and analysis of artificial and real portfolio returns.
- As a Developer my tasks were to design, develop and test two models of portfolio optimization: Markowitz and Black-Litterman.
- Repository: github.com/quatrope/garpar

Continuous Integration of Pyedra **February 2021**

Open Source Project @ IATE, CONICET

- Pyedra is a python library that allows you to fit three different models of asteroid phase functions to your observations.
- My task was to implement a Continuous Integration workflow within the public repository to automatically run the testing suite.
- Repository: github.com/milicolazo/Pyedra

Developer of Arby **November 2020 — July 2021**

Open Source Project @ IATE, CONICET

- Arby is a fully data-driven Python module to construct surrogate models, reduced bases and empirical interpolants from training data.
- My task was to provide time profiling and optimization using Numba JIT compiler.
- Repository: github.com/aaronuv/arby

Back-end Developer of GalaxyChop **November 2021**

Open Source Project @ IATE, CONICET

- GalaxyChop is a Python package that tackles the dynamical decomposition problem of galaxies by using clustering techniques in phase space for stellar galactic components.
- My task was to implement a Fortran backend for the computation of the gravitational force.
- Repository: github.com/vcristiani/galaxy-chop

Postdoctoral Research Scientist **April 2020 — March 2023**

IATE (Institute of Theoretical and Experimental Astronomy), CONICET, Argentina

- Conducted research on the Large Scale Structure of the Universe using gravitational lensing techniques, publishing results in peer-reviewed journals.

Doctoral Researcher **April 2015 — March 2020**

IATE, CONICET, Argentina

- Investigated the physical nature of Galaxy Groups using weak gravitational lensing, resulting in multiple peer-reviewed publications and a PhD in Astronomy.

Scientific Publications in Refereed Journals

- **NIRDust: Probing Hot Dust Emission Around Type 2 AGN Using K-band Spectra** [Software]
  Gaspar, G.; **Chalela, M.**; Cabral, J. B.; Alacoria, J.; Mast, D.; Díaz R. J.
  2024, Monthly Notices of the Royal Astronomical Society. DOI: 10.1093/mnras/stae008.

- **Easy asteroid phase curve fitting for the Python ecosystem: Pyedra** [Software]
  Colazo, M.; Cabral, J. B.; **Chalela, M.**; Sánchez, B. O.
  2022, Astronomy & Computing. DOI: 10.1016/j.ascom.2021.100533.

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- **On the weak-lensing masses of a new sample of galaxy groups**
  Gonzalez, E. J.; Rodriguez, F.; Merchán, M.; García Lambas, D.; Makler, M.; **Chalela, M.**; Pereira, M. E.; Moraes, B.; Shan, H.
  2021, Monthly Notices of the Royal Astronomical Society. DOI: 10.1093/mnras/stab1168.

- **GriSPy: A Python package for Fixed-Radius Nearest Neighbors Search** [Software]
  Chalela, M.; Sillero, E.; Pereyra, L.; García, A. M.; Cabral, J. B.; Lares, M.; Merchán, M.
  2021, Astronomy & Computing. DOI: 10.1016/j.ascom.2020.100443.

- **Measuring the surface mass density ellipticity of redMaPPer galaxy clusters using weak-lensing**
  Gonzalez, E. J.; Makler, M.; García Lambas, D.; **Chalela, M.**; Pereira, M. E.; Van Waerbeke, L.; Shan, H.; Erben, T.
  2021, Monthly Notices of the Royal Astronomical Society. DOI: 10.1093/mnras/staa3570.

- **Setting the scene for BUFFALO: A study of the matter distribution in the HFF galaxy cluster MACS J0416.1-2403 and its parallel field**
  Gonzalez, E. J.; **Chalela, M.**; Jauzac, M.; Eckert, D.; Harvey, D.; Baugh, C. M.; Koekemoer, A.; García Lambas, D.
  2020, Monthly Notices of the Royal Astronomical Society. DOI: 10.1093/mnras/staa745.

- **Compact Groups analysis using weak gravitational lensing II: CFHT Stripe82 data**
  Chalela, M.; Gonzalez, E. J.; Makler, M.; García Lambas, D.; Pereira, M. E. S.; O'Mill, A.; Shan H.
  2018, Monthly Notices of the Royal Astronomical Society. DOI: 10.1093/mnras/sty1537.

- **Compact Groups analysis using weak gravitational lensing**
  Chalela, M.; Gonzalez, E. J.; García Lambas D.; Foëx G.
  2017, Monthly Notices of the Royal Astronomical Society. DOI: 10.1093/mnras/stx242.

- **Weak-lensing measurement of the mass-richness relation using the SDSS data base**
  Gonzalez, E. J.; Rodriguez, F.; García Lambas D.; Merchán, M.; Foëx G.; **Chalela, M.**
  2017, Monthly Notices of the Royal Astronomical Society. DOI: 10.1093/mnras/stw2803.

Selected Talks and Presentations

- **GriSPy: A Python Package for Fixed-Radius Nearest Neighbors** (Poster) 2020
  62nd Annual Meeting of the Argentine Astronomical Association (AAA), virtual.
- **Compact Groups Analysis Using Weak Lensing** (Talk) 2018
  Friends of Friends Meeting, Córdoba, Argentina.
- **A Weak Lensing Analysis of Compact Groups** (Talk) 2017
  Galaxy Groups and Clusters II, La Serena, Chile.
- **A Weak Lensing Analysis of Compact Groups** (Talk) 2017
  Friends of Friends Meeting, Córdoba, Argentina.
- **Compact Groups Analysis Using Weak Lensing** (Talk) 2016
  Latin American Regional IAU Meeting, Cartagena de Indias, Colombia.

Teaching

Software Design for Scientific Computations - Professor **August 2020 — December 2021**
FAMAF (Facultad de Matemática, Astronomía, Física y Computación), UNC, Argentina *Córdoba, Argentina*
Post-Graduate Level Course of 60 hours.

- Unit 1: High-level programming languages (Python).
- Unit 2: Software QA.
- Unit 3: Data persistence.
- Unit 4: Optimization, parallelism, concurrency and high level distributed computing.
- Unit 5: Integrating high with low level languages.
- Unit 6: Packaging and Deployment.

Computing - Student Assistant Professor **November 2010 — June 2012**
FCEFN (Facultad de Ciencias Exactas, Fisicas y Naturales), UNSJ, Argentina *San Juan, Argentina*
Undergraduate Level Course of 60 hours.

- Unit 1: Introduction to coding, pseudo-code.
- Unit 2: Fortran 90.

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Post-Graduate Courses

- **Software Design for Scientific Computing**, taught by Dr. Juan Bautista Cabral.
  2nd semester 2019 (60 hours), FaMAF, U.N.C., Argentina
- **Mathematical Methods of Physics II**, taught by Dr. Guido Raggio.
  1st semester 2017 (120 hours), FaMAF, U.N.C., Argentina
- **Theoretical and Observational Cosmology**, taught by Dr. Diego García Lambas.
  2nd semester 2017 (60 hours), FaMAF, U.N.C., Argentina
- **Numerical Methods in Astrophysics**, taught by Dr. Manuel Merchan.
  1st semester 2015 (60 hours), FaMAF, U.N.C., Argentina
- **Astrostatistics**, taught by Dr. Mariano Dominguez.
  2nd semester 2015 (60 hours), FaMAF, U.N.C., Argentina

Research Visits

- **Ogden Centre for Fundamental Physics**, Durham University, United Kingdom.
  Financed by the EU Horizon2020 RISE project LACEGAL (Latinamerican Chinese European Galaxy Formation Network).
  Research theme: Weak gravitational lensing analysis using HST observational data.
  Adviser: Dr. Mathilde Jauzac.
  Period: 12/09/2018 to 02/16/2019.
- **Centro Brasileiro de Pesquisas Físicas**, Rio de Janeiro, Brasil.
  Research theme: Gravitational lensing analysis using CFHT Stripe 82 Survey.
  Adviser: Dr. Martin Makler.
  Period: 06/28/2017 to 07/28/2017.

Scholarships

- **Post-Doctoral Scholarship**, Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET).
  Adviser: Dr. Diego García Lambas
  Period: april 2020 - march 2023.
- **Doctoral Scholarship**, Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET).
  Adviser: Dr. Carlos Valotto, Co-Adviser: Dr. Diego García Lambas
  Period: april 2015 - march 2020.
- **National Bicentennial Scholarship Program**, FCEFyN, UNSJ, 2011.
- **National Bicentennial Scholarship Program**, FCEFyN, UNSJ, 2012.